



- 6 The curve C has equation $y = \frac{x^2 + a}{x + a}$, where a is a positive constant.

(a) Find the equations of the asymptotes of C . [3]

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(b) Find, in terms of a , the x -coordinates of the stationary points on C . [3]

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(c) Sketch C , stating the coordinates of any intersections with the axes.

[3]

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(d) Sketch the curve with equation $y = \left| \frac{x^2 + a}{x + a} \right|$.

[1]



(e) Find the set of values of a for which $\left| \frac{x^2 + a}{x + a} \right| = a$ has two real solutions. [4]

[illegible]